

COURSE SYLLABUS

Courses Included

S. No.	Course / Subject Name	Department
1	Mathematics I	Department of Mathematics
2	Engineering Mechanics	Department of Mechanical Engineering
3	Career Advancement & Skill Development-I – Communication Skill	Department of English
4	Introduction to Programming (Theory)	Department of Computer Science & Engineering (SOF)
5	Introduction to Programming Using C Lab	Department of Computer Science & Engineering (SOF)
6	Basic Electrical & Electronics Engineering (Theory)	Department of Electronics & Communication Engineering
7	Basic Electrical & Electronics Engineering Lab and Simulation	Department of Electronics & Communication Engineering
8	Basic Computing Lab	Department of Computer Science & Engineering (SOF)
9	Google Tech Fundamentals	School of the Future

Mathematics I

Course Title: Mathematics I	Program: B. Tech. in CSE, CSE-AI, ECE, EE, ME, CE
Year, Semester: 1st Yr., 1st Sem.	Contact Hours/Week: 3-1-0 (L-T-P)
Credits: 4	

Course Objective

1. Analyze and describe the behavior of functions of single and multiple variables, understand sequences and series.
2. Solve systems of linear equations, evaluate eigenvalues and eigenvectors of square matrices.
3. Analyzing differential equations and finding their solutions.

Course Outcome

CO	Course Outcome	Level
CO1	Analyze the behavior and the nature of the curve with calculus of one variable.	K4
CO2	Develop a basic understanding of functions of several variables and their properties.	K4
CO3	Investigate the solutions of system of linear equations using Determinants and Matrices.	K4
CO4	Evaluate Eigen value and vectors of square matrices.	K4
CO5	Examine the nature (viz., convergence, divergence) of sequence and series.	K4
CO6	Analyze differential equations and investigate solutions.	K4

Course Content

Module	Topics	Hours
Module 1	Differential Calculus Differential Calculus (Functions of one variable): Rolle's theorem (statement only), Cauchy's mean value theorem (Lagrange's mean value theorem as a special case), Taylor's and Maclaurin's theorems with remainders, indeterminate forms, concavity and convexity of a curve, points of inflexion, asymptotes and curvature. Differential Calculus (Functions of several variables): Limit, continuity and differentiability of functions of several variables, partial derivatives and their geometrical interpretation, differentials, derivatives of composite and implicit functions, derivatives of higher order and their commutativity, Euler's theorem on homogeneous functions, harmonic functions, Taylor's expansion of functions of several variables, maxima and minima of functions of several variables – Lagrange's method of multipliers.	12 Hours
Module 2	Ordinary Differential Equations Formation of differential equation, First order differential equations - exact, linear and Bernoulli's form, second order differential equations with constant coefficients, method of variation of parameters, general linear differential equations with constant coefficients, Euler's equations, system of differential equations.	10 Hours

Module 3	Sequences and Series Sequences and their limits, convergence of series, comparison test, Ratio test, Root test, Absolute and conditional convergence, alternating series, Power series.	8 Hours
Module 4	Matrix and Determinant Revision of matrix and determinant, rank and nullity, solutions of system of linear equations using Determinants and Matrices; Eigenvalues and eigen vectors, Cayley-Hamilton Theorem, transformation of matrices, adjoint of an operator, normal, unitary, hermitian and skew-hermitian operators, quadratic forms.	15 Hours

Total Lectures: 45 Hours

Books

1. Higher Engineering Mathematics, B. S. Grewal
2. Advanced Engineering Mathematics, Kreyszig
3. A TextBook of Engineering Mathematics, Rajesh Pandey
4. Engineering Mathematics, B. K. Pal, K. Das

Engineering Mechanics

Course Title: Engineering Mechanics	Program: B. Tech. in Mechanical Engineering
Year, Semester: 1st Yr., 2nd Sem.	Contact Hours/Week: 3-0-0 (L-T-P)
Credits: 3	

Course Objective

- Understand the basics of vector mechanics and its applications in engineering mechanics.
- Analyze problems in statics.
- Analyze problems in dynamics of particles.

Course Outcome

CO	Course Outcome	Level
CO1	To understand the basics of vector mechanics and its application in engineering mechanics.	K2
CO2	To understand different force systems and the methods of finding their resultants and to be well-versed with the conditions of equilibrium in 2D.	K2
CO3	To be able to apply the laws of static equilibrium in solving problems and perform analysis of statically determinate trusses.	K4
CO4	To be able to compute centroids of plane areas, composite areas and to be able to compute area moments of inertias and radii of gyration of plane figures.	K3
CO5	To understand basic principles of kinematics of particles, plane, rectilinear and curvilinear coordinate systems and projectile motion.	K3
CO6	To understand basic principles of kinetics of particles leading to Newton's laws and to be able to apply the work-energy and the linear impulse-linear momentum theorems in solving typical problems.	K3

Course Content

Module	Topics	Hours
Module 1	Introduction Fundamentals of Mechanics: Introduction to mechanics; Basic concepts – mass, space, time and force; Particles and rigid bodies; Scalars and vectors; Free, sliding, fixed and unit vectors; Addition, subtraction and multiplication of two vectors; scalar triple product and vector product of 3 vectors.	4 Hours
Module 2	Force Systems and Equilibrium Force systems: Introduction to different force systems; Composition of forces – triangle, parallelogram and polygon law of forces, and addition of two parallel forces; Resolution of forces; Moment of a force, Varignon's theorem; Couples; Force-couple system; Resultant of a force system. Equilibrium: Free body diagram, equilibrium conditions in 2 dimensions, equilibrium of systems involving friction.	9 Hours
Module 3	Structures Plane Truss: Statically determinate trusses; Force analysis of a truss - method of joints, method of sections.	5 Hours

Module 4	Distributed Forces Line, area and volume distributions of forces; Centre of gravity; Centre of mass; Centroids of plane figures; Centroids of composite areas. Moment of Inertia: Area moment of inertia; Perpendicular and Parallel axes theorems pertaining to moment of inertia; Radius of gyration.	7 Hours
Module 5	Kinematics of Particles Differential equations of kinematics – plane, rectilinear and curvilinear motions; Cartesian co-ordinate system; Normal and tangent co-ordinate system, projectile motion.	8 Hours
Module 6	Kinetics of Particles Newton's second law of motion; Work and energy principle – gravitational potential energy, elastic potential energy, kinetic energy, power, work-energy theorem, principle of impulse and momentum.	12 Hours

Total Lectures: 45 Hours

Books

1. J. L. Meriam and L. G. Kraige, Engineering Mechanics (Vol.1 & 2), Wiley India, 2017.
2. Shames I. H., Rao G. K. M., Engineering Mechanics, Pearson, 2005.
3. Khurmi R.S., A Textbook of Engineering Mechanics, S. Chand, 2018.
4. Bhavikatti S. S, Engineering Mechanics, New Age International Publishers, 2021.

Career Advancement & Skill Development-I – Communication Skill

Course Title: Career Advancement & Skill Development-I – Communication Skill	Program: B. Tech.
Year, Semester: 1st Yr., 1st Sem.	Contact Hours/Week: 2-0-0 (L-T-P)
Credits: 2	

Course Objective

7. Develop English proficiency for clear, precise, and confident workplace communication.
8. Enhance practical skills in vocabulary, grammar, pronunciation, speaking, and writing.
9. Apply communication theories to improve professional and interpersonal interactions.

Course Outcome

CO	Course Outcome	Level
CO1	Explain fundamental communication principles and their relevance in workplace interactions.	K2
CO2	Apply grammar and language skills to construct precise and coherent spoken and written communication.	K3
CO3	Demonstrate fluency in spoken English through pronunciation drills, vocabulary building, and interactive conversations.	K4
CO4	Construct well-organized sentences, paragraphs, and linked paragraphs to enhance professional writing.	K3
CO5	Develop and revise written communication by employing strategies for drafting, editing, and proofreading.	K3
CO6	Assess and refine communication skills to ensure clarity, precision, and confidence in workplace interactions.	K4

Course Content

Module	Topics	Hours
Module 1	Introduction to Communication Definition of Communication, Importance of Communication in the Workplace, Introduction to Communication Theory, Elements of Effective Communication, Barriers to Communication, Verbal and Non-Verbal Communication, Role of Culture in Communication.	5 Hours
Module 2	Language and Grammar Skills Fundamentals of English Grammar, Sentence Structure and Syntax, Parts of Speech, Tenses and their Usage, Common Errors in Grammar, Punctuation and Mechanics, Effective Use of Vocabulary, Word Formation and Usage, Formal vs. Informal Language.	5 Hours
Module 3	Speaking Skills Principles of Effective Speaking, Pronunciation Drills, Sounds of English: Vowels and Consonants, Stress and Intonation, Developing Conversational Skills, Speaking with Clarity and Confidence, Public Speaking Basics, Expressing Opinions and Arguments, Active Listening and Response.	5 Hours

Module 4	Writing Skills The Writing Process: Planning, Drafting, Revising, Editing, Writing Effective Sentences and Paragraphs, Paragraph Development and Coherence, Formal and Informal Writing Styles, Writing Emails and Workplace Documents, Writing Reports and Memos, Common Writing Errors and How to Avoid Them.	5 Hours
Module 5	Practical Language Application Building Vocabulary through Context, Word Choice and Precision, Constructing Grammatically Correct Sentences, Exercises in Sentence Formation, Pronunciation Drills and Accent Neutralization, Role-Plays and Dialogues, Group Discussions and Debates, Writing and Structuring Paragraphs, Linking Paragraphs for Coherent Writing.	5 Hours
Module 6	Professional Communication in the Workplace Workplace Communication Etiquette, Business Correspondence, Writing Professional Emails, Preparing Presentations, Communicating in Meetings, Handling Workplace Conversations, Persuasive and Negotiation Skills, Overcoming Communication Barriers, Strategies for Effective Workplace Communication.	5 Hours

Total Lectures: 30 Hours

Books

1. Sanjay Kumar, Pushp Lata, "Communication Skills", Oxford University Press, 2015, ISBN: 9780199457069
2. M Ashraf Rizvi, "Effective Technical Communication", McGraw Hill Education, 2017, ISBN 9352606108
3. Steven A. Beebe, Susan J. Beebe, and Mark V. Redmond, "Interpersonal Communication: Relating to Others", Pearson, 2013.
4. Judee K. Burgoon, Laura K. Guerrero, and Kory Floyd, "Nonverbal Communication", Routledge, 2016.
5. Ronald B. Adler, Lawrence B. Rosenfeld, and Russell F. Proctor II, "Interplay: The Process of Interpersonal Communication", Oxford University Press, 2017.
6. Joseph A. DeVito, "The Interpersonal Communication Book", Pearson, 2015.
7. Sarah Trenholm and Arthur Jensen, "Interpersonal Communication", Oxford University Press, 2013.
8. John Stewart, "Bridges Not Walls: A Book About Interpersonal Communication", McGraw-Hill Education, 2011.
9. Pamela J. Kalbfleisch, "Interpersonal Communication: Evolving Interpersonal Relationships", Routledge, 2013.
10. Mark L. Knapp, John A. Daly, and Frederick P. M. Boster, "Interpersonal Communication Handbook", Sage Publications, 2011.

Introduction to Programming

Course Title: Introduction to Programming	Program: B. Tech. in CSE-AI
Year, Semester: 1st Yr., 1st Sem.	Contact Hours/Week: 3-0-0 (L-T-P)
Credits: 3	

Course Objective

10. Develop algorithmic problem-solving skills and implement them in C programs.
11. Apply modular programming, recursion, and data structures to create interactive C programs.
12. Utilize advanced C concepts like structures, pointers, and linked lists for efficient programming.

Course Outcome

CO	Course Outcome	Level
CO1	Analyze algorithmic solutions to problems.	K4
CO2	Construct algorithms using C programming.	K3
CO3	Apply interactive input/output, arithmetic expressions, repetitions, decision-making, and arrays in programs.	K3
CO4	Organize modular C programs using functions, including recursion.	K3
CO5	Categorize programs using structures, unions, pointers, and linked lists.	K4
CO6	Utilize file input and output operations in programs.	K3

Course Content

Module	Topics	Hours
Module 1	Introduction to C Language Character set, Variables and Identifiers, Built-in Data Types, Variable Definition, Arithmetic operators and Expressions, Constants and Literals, Simple assignment statement, Basic input/output statement, Simple 'C' programs.	4 Hours
Module 2	Conditional Statements and Loops Decision making within a program: Conditions, Relational Operators, Logical Connectives, if statement, if-else statement. Loops: while loop, do while, for loop, Nested loops, Infinite loops, switch statement, Structured Programming.	6 Hours
Module 3	Arrays One dimensional arrays: Array manipulation, Searching, Insertion, and Deletion of an element from an array, finding the largest / smallest element in an array; Two dimensional arrays, Addition/multiplication of two matrices, transpose of a square matrix, Null terminated strings as array of characters, Representation of sparse matrices.	6 Hours
Module 4	Functions Top-down approach of problem solving; Modular programming and functions; Standard Library of C functions; Prototype of a function: Formal parameter list, Return Type, Function call, Block structure; Passing arguments to a Function: Call by reference, Call by value, Recursive Functions, Arrays as function arguments.	7 Hours

Module 5	Structures and Unions Structure variables, Initialization, Structure assignment, Nested structure, Structures and Functions, Structures and arrays: Arrays of structures, Structures containing arrays, Unions.	5 Hours
Module 6	Pointers Address operators, Pointers type declaration, Pointer assignment, Pointer initialization, Pointer arithmetic, Functions and pointers, Arrays and Pointers, Pointer arrays.	9 Hours
Module 7	Self-Referential Structures and Linked Lists Creation of a singly connected linked list, traversing a linked list, Insertion into a linked list, Deletion from a linked list.	3 Hours
Module 8	File Processing Concept of Files, File opening in various modes and closing of a file, Reading from a file, writing onto a file.	5 Hours

Total Lectures: 45 Hours

Books

1. B W Kernighan and D.M. Ritchie, The C Programming Language, Prentice Hall of India.
2. K. Venugopal and Sudeep R Prasad, Programming with C, McGraw Hill.
3. R G Dromey, How to Solve it by Computer, Prentice Hall of India.
4. Jones, Robin and Stewart, The Art of C Programming, Narosa Publishing House.
5. A Kenneth, C Problem Solving and Programming, Prentice Hall International.
6. H. Schildt, C: The Complete Reference, 4th Edition, McGraw Hill.

Introduction to Programming Using C Lab

Course Title: Introduction to Programming Using C Lab	Program: B. Tech. in CSE-DS
Year, Semester: 2nd Yr., 1st Sem.	Contact Hours/Week: 0-0-2 (L-T-P)
Credits: 1	

Course Objective

- 13.To familiarize students with the fundamentals of the C programming language, including its character set, data types, operators, and basic input/output constructs.
- 14.To develop the ability to design algorithms and write structured C programs using decision-making statements, loops, arrays, and strings.
- 15.To build proficiency in modular programming through functions, and in organizing data using structures, unions, and pointers.
- 16.To provide hands-on experience with dynamic data structures such as linked lists, and with file handling operations in C.

Course Outcome

CO	Course Outcome	Level
CO1	Identify the character set, data types, operators, and basic input/output statements used in C, and write simple C programs.	K2
CO2	Apply decision-making statements, loops, and the switch statement to develop structured C programs.	K3
CO3	Design and manipulate one- and two-dimensional arrays and character strings, including searching, insertion, deletion, and matrix operations.	K3
CO4	Develop modular programs using user-defined and library functions, with different parameter-passing techniques including recursion.	K3
CO5	Use structures, unions, and arrays of structures to represent and manipulate composite data.	K3
CO6	Apply pointers for memory referencing, pointer arithmetic, and interaction with functions and arrays.	K3
CO7	Implement singly linked lists with creation, traversal, insertion, and deletion operations.	K4
CO8	Perform file handling operations in C, including opening, reading, writing, and closing files.	K3

Course Content

Module	Topics	Hours
Module 1	Introduction to C Programming Character set, Variables and Identifiers, Built-in Data Types, Variable Definition, Arithmetic operators and Expressions, Constants and Literals, Simple assignment statement, Basic input/output statement, Simple 'C' programs.	6 Hours

Module 2	Decision Making and Looping Decision making within a program: Conditions, Relational Operators, Logical Connectives, if statement, if-else statement. Loops: while loop, do while loop, for loop, Nested loops, Infinite loops, switch statement, Structured Programming.	6 Hours
Module 3	Arrays and Strings One dimensional arrays: Array manipulation, Searching, Insertion, and Deletion of an element from an array, finding the largest/smallest element in an array; Two dimensional arrays, Addition/multiplication of two matrices, transpose of a square matrix, Null terminated strings as array of characters, Representation of sparse matrices.	6 Hours
Module 4	Functions and Modular Programming Top-down approach of problem solving; Modular programming and functions; Standard Library of C functions; Prototype of a function: Formal parameter list, Return Type, Function call, Block structure; Passing arguments to a Function: Call by reference, Call by value, Recursive Functions, Arrays as function arguments.	6 Hours
Module 5	Structures and Unions Structure variables, Initialization, Structure assignment, Nested structure, Structures and Functions, Structures and arrays: Arrays of structures, Structures containing arrays, Unions.	4 Hours
Module 6	Pointers Address operators, Pointer type declaration, Pointer assignment, Pointer initialization, Pointer arithmetic, Functions and pointers, Arrays and Pointers, Pointer arrays.	4 Hours
Module 7	Linked Lists Creation of a singly connected linked list, traversing a linked list, Insertion into a linked list, Deletion from a linked list.	4 Hours
Module 8	File Handling Concept of Files, File opening in various modes and closing of a file, Reading from a file, writing onto a file.	4 Hours

Total Lab Hours: 40 Hours

Basic Electrical & Electronics Engineering

Course Title: Basic Electrical & Electronics Engineering	Program: B. Tech. SOF
Year, Semester: 1st Yr., 2nd Sem.	Contact Hours/Week: 3-1-0 (L-T-P)
Credits: 4	

Course Objective

17. Analyze and describe the basic electrical quantities, circuit elements, and their voltage-current relationships.
18. Design and analyze diode circuits, transistor biasing, and operational amplifier applications.
19. Understand the operation and characteristics of semiconductor devices like diodes, BJTs, JFETs, and MOSFETs.
20. Analyzing differential working principles of single-phase transformers, including voltage transformation and regulation.

Course Outcome

CO	Course Outcome	Level
CO1	Apply fundamental electrical concepts to solve basic electrical engineering problems.	K3
CO2	Analyze DC electrical networks using appropriate circuit analysis techniques.	K4
CO3	Analyze and understand AC circuits and power systems.	K4
CO4	Explain the characteristics and operating principles of semiconductor devices and their applications.	K2
CO5	Apply the operating principles of semiconductor devices to analyze their use in electronic circuits.	K3
CO6	Explain the operating principles and applications of transformers.	K2

Course Content

Module	Topics	Hours
Module 1	Introduction Basic electrical quantities, Voltage, Current, Power. Basic Electrical elements: Resistance, Inductance, Capacitance. Their voltage-current relationship. Voltage and current sources.	4 Hours
Module 2	DC Network Analysis KCL and KVL and their applications in purely resistive circuits. Concept of linear, bilateral networks. Source conversion, Star-Delta conversion.	6 Hours
Module 3	DC Network Theorems Superposition Theorem, Thevenin's Theorem, Norton's Theorem, Maximum Power Transfer Theorem.	8 Hours
Module 4	Sinusoidal Steady State Analysis Fundamentals of AC circuits, RMS, Average, Form factor, resistive, inductive and capacitive circuits, Active, reactive and apparent power calculation, Series and parallel resonance, quality factor.	6 Hours

Module 5	Semiconductor Devices Energy bands in solids. Intrinsic and extrinsic semiconductors. P-N junctions. Semiconductor diodes: Zener and Varactor diodes. Bipolar transistors (operation, characteristics).	4 Hours
Module 6	Electronic Circuits Diode Circuits, BJT biasing & Operation of JFET, MOSFET.	7 Hours
Module 7	OPAMPs Properties of an ideal and a practical OPAMP. Block diagram. Concept of Virtual Short, Inverting and Non-inverting amplifiers, Summing and Differencing amplifier, Differentiator and Integrator.	5 Hours
Module 8	1-Ph Transformers Faraday's Law, EMF generation (dynamic and static), B-H curve, Construction and operation of single phase transformer: voltage and current transformation, no-load operation, voltage regulation on resistive load.	5 Hours

Total Lectures: 45 Hours

Books

1. Nagrath and Kothari, Basic Electrical and Electronics Engineering, Tata McGraw-Hill

Basic Electrical & Electronics Engineering Lab and Simulation

Course Title: Basic Electrical & Electronics Engineering Lab and Simulation	Program: B. Tech. SOF
Year, Semester: 1st Yr., 2nd Sem.	Contact Hours/Week: 0-0-3 (L-T-P)
Credits: 1.5	

Course Objective

21. Introduce fundamental electrical and electronic circuit theorems and develop analytical skills for solving electrical networks.
22. Familiarize students with essential circuit components, including R-L-C circuits, diodes, rectifiers, and fluorescent lamps, and their practical applications.
23. Enhance hands-on laboratory skills by conducting experiments on circuit analysis, diode characteristics, and rectifier efficiency evaluation.

Course Outcome

CO	Course Outcome	Level
CO1	Identify and understand fundamental electrical and electronic circuit theorems and their applications.	K1
CO2	Explain the working principles of R-L-C circuits, diodes, rectifiers, and fluorescent lamps.	K2
CO3	Apply circuit theorems such as Superposition and Thevenin's Theorem to analyze electrical networks.	K3
CO4	Conduct experiments to measure and analyze V-I characteristics of P-N junction and Zener diodes.	K3
CO5	Evaluate the efficiency and power factor of electrical circuits, rectifiers, and fluorescent lamps.	K4
CO6	Compare different rectifier circuits and analyze their output waveforms and ripple factors.	K4

Course Content

Module	Topics	Hours
Exp 1	Study and Identification of Electrical Components, Measuring Instruments and Safety Devices To identify and understand the functions of various basic electrical components, measuring instruments and safety devices, and to learn their safe and correct use in electrical laboratory applications.	5 Hours
Exp 2	Verification of Superposition Theorem Theoretical foundation of superposition theorem, Application in linear electrical circuits, Step-by-step circuit analysis with multiple voltage/current sources, Practical applications in circuit design, troubleshooting, and network analysis.	5 Hours

Exp 3	Study of R-L-C Series Circuit Characteristics of resistance (R), inductance (L), and capacitance (C) in AC circuits, Impedance (Z) and phase angle, Voltage and current phase relationships, Leading and lagging power factor, Practical applications in circuit analysis and troubleshooting.	5 Hours
Exp 4	Verification of Thevenin's Theorem Theoretical foundation of Thevenin's theorem, Converting complex circuits into Thevenin equivalent, Measuring Thevenin voltage (V_{th}) and resistance (R_{th}), Practical applications in circuit design and network analysis.	5 Hours
Exp 5	Characteristics of Fluorescent Lamp Gas discharge and phosphor coating in light production, Role of starter, choke (ballast), and electrodes, Measuring voltage, current, and power consumption, Efficiency comparison with incandescent and LED lamps, Impact of inductive ballast on power factor and improvement methods, Performance comparison of electromagnetic vs. electronic ballasts, Energy savings, lifespan, and environmental concerns (mercury content).	5 Hours
Exp 6	Familiarization with Basic Electronic Components Identification, specifications, and testing of R, L, and C components (Color codes), Potentiometers, switches (SPDT, DPDT, DIP), Breadboards and Printed Circuit Boards (PCBs), Active components: Diodes, BJTs, JFETs, MOSFETs, Power transistors, SCRs, LEDs.	5 Hours
Exp 7	Study of V-I Characteristics of P-N Junction Diode in Forward Bias Depletion layer and barrier potential, Forward bias operation, Breakdown voltage and Peak Inverse Voltage (PIV), Knee voltage and ideal PN junction diode characteristics.	5 Hours
Exp 8	V-I Characteristics of Zener Diode in Reverse Bias Depletion layer and barrier potential, Reverse bias operation, Breakdown voltage and Peak Inverse Voltage (PIV), Knee voltage and ideal Zener diode characteristics.	5 Hours
Exp 9	Study of Half-Wave and Full-Wave Rectifier Half-wave and full-wave rectifiers (Center-tap and Bridge), Output waveforms and voltage regulation, Ripple factor and rectifier efficiency.	5 Hours

Total Lab Hours: 45 Hours

Books

1. Boylestad, R. L., & Nashelsky, L. (2015). Electronic Devices and Circuit Theory (11th ed.). Pearson.
2. Hayt, W. H., Kemmerly, J. E., & Durbin, S. M. (2018). Engineering Circuit Analysis (9th ed.). McGraw-Hill Education.
3. Sedra, A. S., & Smith, K. C. (2016). Microelectronic Circuits (7th ed.). Oxford University Press.
4. Malvino, A. P., & Bates, D. J. (2016). Electronic Principles (8th ed.). McGraw-Hill Education.

Basic Computing Lab

Course Title: Basic Computing Lab	Program: B. Tech. in CSE-DS
Year, Semester: 1st Yr., 1st Sem.	Contact Hours/Week: 0-0-2 (L-T-P)
Credits: 1	

Course Objective

24. To introduce students to the UNIX/Linux environment and familiarize them with fundamental system operations, commands, and file management techniques.
25. To develop proficiency in shell scripting and command-line utilities for automating tasks, managing processes, and handling files efficiently.
26. To provide hands-on experience with GitHub operations and debugging techniques while enhancing students' ability to work with text processing tools, redirection, and file compression in a UNIX/Linux environment.

Course Outcome

CO	Course Outcome	Level
CO1	Be familiar with the UNIX/Linux operating system.	K2
CO2	Develop proficiency in using shell commands and writing basic shell scripts.	K3
CO3	Understand file systems, process management, and user permissions.	K2
CO4	Understand basic GitHub operations and debugging of programs.	K3
CO5	Apply fundamental text processing tools and commands such as grep, find, and text editors (vi/nano) for efficient file manipulation and searching.	K4
CO6	Utilize redirection, piping, and file compression techniques to manage data effectively in a UNIX/Linux environment.	K4

Course Content

Module	Topics	Hours
Module 1	Introduction to UNIX/Linux and Basic Commands Overview of UNIX/Linux operating systems, Logging into UNIX/Linux systems, Basic system commands: ls, cd, pwd, cp, mv, rm, clear, man, who, date, cal, etc. Understanding the file system hierarchy: /, /home, /bin, /usr, /var, etc.	6 Hours
Module 2	File and Process Management File and Directory Management: Creating, removing, and organizing files and directories, Commands: mkdir, rmdir, touch, chmod, chown, rm, find, etc. Understanding file permissions and ownership (rwx permissions, chmod command). Process Management: Viewing active processes (ps, top, htop), Controlling processes: kill, bg, fg, jobs, nice, and renice, Understanding process states: running, sleeping, zombie.	6 Hours
Module 3	Text Processing and Basic Shell Scripting Text Editors (vi, nano): Creating, editing, saving, and exiting files, Working with commands like grep, cat, more, less, sed, and awk. Basic Shell Scripting: Writing simple shell scripts (bash), Understanding variables, loops (for, while), and conditional statements (if, elif, else), Creating automation scripts for file operations and system monitoring.	6 Hours

Module 4	Redirection, Piping, and File Compression Redirection and Piping: Input/output redirection (>, >>, <), Piping () for command chaining. File Compression and Archiving: Working with gzip, tar, zip, unzip, Creating and extracting archives for data backup.	6 Hours
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Total Lab Hours: 24 Hours

Teaching Plan

Module No.	Exp. No.	Topics to be Covered	CO	K-Level
Module 1: Introduction to UNIX/Linux and Basic Commands	1	Log into a UNIX/Linux system and explore basic system information commands: who, date, cal.	CO1	K2
Module 1: Introduction to UNIX/Linux and Basic Commands	2	Practice navigation commands with options: pwd, ls (-l, -a, -lt), cd.	CO1	K2
Module 1: Introduction to UNIX/Linux and Basic Commands	3	Perform basic file operations using cp, mv, rm, and clear commands.	CO1	K2
Module 1: Introduction to UNIX/Linux and Basic Commands	4	Explore the UNIX/Linux file system hierarchy (/ , /home, /bin, /usr, /var) using the man command for reference.	CO1	K2
Module 1: Introduction to UNIX/Linux and Basic Commands	5	Write and execute a sequence of commands to display detailed system and user information.	CO1	K2
Module 1: Introduction to UNIX/Linux and Basic Commands	6	Practice combining basic commands with multiple options to perform composite file-listing tasks.	CO1	K2
Module 2: File and Process Management	7	Create, remove, and organize directories and files using mkdir, rmdir, and touch.	CO3	K2
Module 2: File and Process Management	8	Set and modify file permissions and ownership using chmod and chown; interpret rwx permissions.	CO3	K2
Module 2: File and Process Management	9	Search for files and directories using the find command with various options.	CO3	K2
Module 2: File and Process Management	10	Monitor active processes using ps, top, and htop.	CO3	K2
Module 2: File and Process Management	11	Control running processes using kill, bg, fg, and jobs.	CO3	K2
Module 2: File and Process Management	12	Adjust process scheduling priority using nice and renice; identify process states (running, sleeping, zombie).	CO3	K2
Module 3: Text Processing and Basic Shell Scripting	13	Create, edit, and save a text file using the vi editor.	CO5	K4

Module 3: Text Processing and Basic Shell Scripting	14	Create, edit, and save a text file using the nano editor.	CO5	K4
Module 3: Text Processing and Basic Shell Scripting	15	Perform text searching and viewing operations using grep, cat, more, and less.	CO5	K4
Module 3: Text Processing and Basic Shell Scripting	16	Perform pattern-based text processing using sed and awk.	CO5	K4
Module 3: Text Processing and Basic Shell Scripting	17	Write a shell script to declare variables and perform basic input/output operations.	CO2	K3
Module 3: Text Processing and Basic Shell Scripting	18	Write a shell script using a for loop to automate a repetitive file operation.	CO2	K3
Module 3: Text Processing and Basic Shell Scripting	19	Write a shell script using a while loop and if-elif-else conditional statements.	CO2	K3
Module 3: Text Processing and Basic Shell Scripting	20	Write a shell script to automate system monitoring (e.g., disk usage, memory usage) and log the output to a file.	CO2	K3
Module 4: Redirection, Piping, File Compression and Version Control	21	Demonstrate input/output redirection using >, >>, and < operators.	CO6	K4
Module 4: Redirection, Piping, File Compression and Version Control	22	Demonstrate command chaining using the pipe () operator with multiple commands.	CO6	K4
Module 4: Redirection, Piping, File Compression and Version Control	23	Create and extract compressed archives using gzip and tar.	CO6	K4
Module 4: Redirection, Piping, File Compression and Version Control	24	Create and extract compressed archives using zip and unzip for data backup.	CO6	K4
Module 4: Redirection, Piping, File Compression and Version Control	25	Create a GitHub repository and practice the clone operation to set up a local copy.	CO4	K3
Module 4: Redirection, Piping, File Compression and Version Control	26	Practice the GitHub add, commit, push, and pull workflow, and apply basic debugging techniques to a sample program.	CO4	K3

Google Tech Fundamentals

Course Title: Google Tech Fundamentals	Program: School of the Future
Year, Semester: Semester I	Contact Hours/Week: 3 (Theory) — L-T-P: 3-0-0
Credits: 3	

Course Objective

- 27.To introduce foundational concepts of computing, internet architecture, and data storage.
- 28.To develop working familiarity with Google Workspace and cloud-based tools as instruments of learning.
- 29.To build an accurate conceptual understanding of how everyday consumer technology functions.
- 30.To cultivate informed, safe, and responsible digital habits among first-semester students.

Course Outcome

CO	Course Outcome	Level
CO1	Explain fundamental hardware, software, and file management concepts.	-
CO2	Describe the architecture and working principles of the internet.	-
CO3	Analyse the internal working of mobile and web applications.	-
CO4	Differentiate data storage models and cloud storage principles.	-
CO5	Evaluate how search, recommendation, and tracking systems shape user experience.	-
CO6	Apply digital safety, privacy, and security principles in everyday technology use.	-

Course Content

Module	Topics	Hours
Module 1	Computer Fundamentals Central Processing Unit (CPU), Random Access Memory (RAM), storage devices, input and output devices; Hardware versus software; Files, folders, file formats, and file size units (bits to gigabytes); Illustrative case discussion: battery charging behaviour, why battery percentage does not drain linearly, and perceived device slowdown over time; Why more megapixels doesn't mean a better photo; How a touchscreen actually detects your finger; Why 90Hz/120Hz screens feel smoother (refresh rate).	9 Hours
Module 2	Internet and How It Works Internet versus World Wide Web; browsers, websites, and servers; IP addresses, Domain Name System (DNS), Uniform Resource Locators (URLs); HTTP/HTTPS protocols; Wi-Fi versus mobile data; bandwidth and connection speed; Case study: the request–response cycle of opening a website; Adaptive video streaming: why playback quality adjusts to available bandwidth; Network connectivity resets: why toggling airplane mode can restore a lost signal.	8 Hours
Module 3	Inside a Mobile and Web Application Client–server architecture: what occurs when an application is launched; Software updates and application crashes; Software bugs: causes and classification; Offline functionality and application size; Application permissions: camera, microphone, and location access, and the effect of denying them.	7 Hours

Module 4	Data Storage and File Systems File compression and file formats; Local versus cloud storage; the storage hierarchy; File deletion and recovery mechanisms, including why uninstalling an application does not always free all the storage it used; Data centre architecture and cloud storage infrastructure; What "cache" is and why clearing it sometimes fixes problems.	7 Hours
Module 5	Search, Recommendation, and Tracking Systems Search engine ranking principles; Recommendation and content feed algorithms; Cookies, web tracking, and targeted advertising; Cross-application tracking: how identifiers follow users across different apps; Voice assistants and wake-word detection: how always-on listening for a trigger phrase works without continuous recording.	7 Hours
Module 6	Digital Safety, Privacy, and Security Password security and authentication mechanisms; Encryption fundamentals; two-factor authentication; Phishing and social engineering: case studies; Digital footprint; overview of the Digital Personal Data Protection (DPDP) Act, 2023; End-to-end encryption in messaging applications: what is and is not protected; Malware and viruses: common infection vectors on personal devices.	7 Hours

Total Theory Contact Hours: 45

Evaluation Scheme

Component	Weightage (Marks)
Internal Assessment (Class Tests, Assignments, Attendance)	20
End Semester Examination (Theory)	80
Total	100